

Assignment 3

(Sep. 28-Oct. 5)

Problem 1:

For each of the following linear programs determine whether it is **infeasible**, **unbounded**, or if it has an **optimal point**. In each case, justify your answer (e.g. with a certificate that the linear program has this property).

(a)

$$\begin{array}{ll}\min_{x_1, x_2} & x_1 + 2x_2 \\ \text{s.t.} & -x_1 \leq 0, \\ & -x_2 \leq 0, \\ & -x_1 - x_2 \leq -1.\end{array}$$

(b) Consider

$$\begin{array}{ll}\min_{x_1, x_2} & -x_1 - x_2 \\ \text{s.t.} & x_1 \leq 2, \\ & x_2 \leq 3, \\ & -x_1 \leq 0, \\ & -x_2 \leq 0.\end{array}$$

(c)

$$\begin{array}{ll}\min_{x_1, x_2} & -x_1 + x_2 \\ \text{s.t.} & -x_1 \leq 0, \\ & -x_2 \leq 0.\end{array}$$

(d)

$$\begin{array}{ll}\min_{x_1, x_2} & x_1 + x_2 \\ \text{s.t.} & x_1 \leq 0, \\ & -x_1 \leq -1.\end{array}$$

Problem 2:

A company has a set J of jobs to be assigned to a set I of employees. Employee $i \in I$ needs c_{ij} hours to complete job $j \in J$. The company wants to complete all jobs by assigning exactly one job to each employee, while minimizing the total number of hours spent to complete all jobs. For this reason, the company has formulated the following IP:

$$\begin{aligned} \min \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} \\ \sum_{i \in I} x_{ij} &= 1 \quad \forall j \in J \\ \sum_{j \in J} x_{ij} &= 1 \quad \forall i \in I \\ x &\geq \mathbf{0}, \text{ integer} \end{aligned}$$

Consider the LP obtained by the above IP after ignoring the integrality constraints, that is:

$$\begin{aligned} \min \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} \\ \sum_{i \in I} x_{ij} &= 1 \quad \forall j \in J \\ \sum_{j \in J} x_{ij} &= 1 \quad \forall i \in I \\ x &\geq \mathbf{0}, \end{aligned}$$

Show that the above LP is feasible (i.e., that it admits at least one feasible solution) if and only if $|I| = |J|$.

Note: to show that the LP is feasible, you need to provide one feasible solution. To show that the LP is infeasible, provide a certificate of infeasibility as in Proposition 2.1 (p. 46) in the textbook.

Problem 3:

(a) Here is a linear program

$$\begin{array}{ll}\text{minimize} & 3x_1 + 2x_2 + x_3 \\ \text{subject to} & x_1 + x_2 + x_3 \leq 8, \\ & 2x_1 - x_2 + x_3 \geq 2, \\ & x_1 + 3x_2 - x_3 \leq 10, \\ & -x_1 + x_2 + 2x_3 \geq 1. \quad x_1, x_2, 2x_3 \geq 0.\end{array}$$

Write an equivalent linear program in standard equality form.

(b) Consider the following matrix (where the top row just numbers the columns).

$$A = \begin{array}{ccccc} & 1 & 2 & 3 & 4 & 5 \\ \begin{pmatrix} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} & \end{array}.$$

which of the following sets of columns are bases of this matrix?

$$\{1, 2, 5\}, \{1, 2\}, \{1, 2, 3\}, \{1, 3, 5\}$$

(c) Consider the same matrix as in (b) in the equation

$$Ax = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Put this linear equation in canonical form, using the basis $\{1, 2, 4\}$.